Cambridge Instruments & Engg. Co.

EST. 1963 · HOWRAH, INDIA

INSULATION TESTER · MOTOR-CUM-HAND DRIVEN

CIE/777 HM

Mains motor with hand-crank backup

Direct-reading insulation resistance testing

A universal, multi-purpose instrument for checking insulation and conductance under simulated operating conditions in high-tension electrical and electronic fields. The dielectric absorption test needs the voltage applied for several minutes, and driving the instrument by motor makes this easy. Polarisation index is the resistance after 10 minutes divided by the resistance after 1 minute, and better results come from longer test periods. The high sensitivity of the instrument gives sensible readings in these tests, and it can run for hours when locating intermittent shorts, defective electrical connections and insulation or conductor failure. The 777 HM can also be driven by hand, for use where mains power is not available.



CIE/777 HM · ILLUSTRATIVE IMAGE

14	5	3 s
STANDARD RANGES	RATED VOLTAGES	RESPONSE TIME

STANDARD RATED TEST VOLTAGES

500 V | 1,000 V | 2,500 V | 5,000 V | 10,000 V

01 HIGHLIGHTS

- ✓ Operated from 230 V, 50 Hz A.C. mains
- ✓ Collapsible generator handle
- ✓ Seasoned wooden cabinet with sunmica finish, detachable lid, carrying handle and lock
- ✓ Highly sensitive cross-coil indicator
- ✓ Externally accessible infinity adjuster
- ✓ Terminals accept 4 mm plug and spade terminations
- ✓ Conforms to IS 2992:1987
- ✓ Can also be operated by hand crank
- ✓ Centrifugal clutch (governor) for constant output voltage in hand mode
- ✓ Anti-parallax mirror scale with knife-edge pointer
- ✓ Guard ring around the line terminal, plus a guard terminal
- ✓ ON-OFF switch, jewel lamp for mains ON, safety fuse
- ✓ Accurate, quick response (3 seconds)

02 TYPICAL APPLICATIONS

- Field insulation testing where mains supply is uncertain or unavailable
- Sustained PI and dielectric absorption testing (motor-driven mode)
- Remote substation and railway traction equipment commissioning
- HV transformer and cable insulation assessment

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03 HOW IT WORKS

PRINCIPLE

A mains-driven motor, or a hand crank, turns the generator, and a static rectifier supplies the D.C. test voltage to a cross-coil ohm meter.

CONSTRUCTION

Housed in the same seasoned wooden cabinet as the CIE/777, with sunmica finish, detachable lid, carrying handle and lock. A foldable hand-generator mechanism adds weight: 12 kg including top cover, against 9 kg for the CIE/777.

04 SALIENT FEATURES

Mains motor drive

A built-in motor for 230 V, 50 Hz A.C. mains, with a static rectifier and a cross-coil ohm meter. The test voltage can be applied for as long as the test needs.

Guard ring and guard terminal

A guard ring around the line terminal and a guard terminal eliminate errors due to surface leakage. An externally accessible infinity adjuster is also provided.

Safe, simple controls

A highly insulated panel carries an ON-OFF switch, a jewel lamp indicating mains ON and a safety fuse protecting the motor, with a 3-pin socket and a 2 m supply cord.

Hand-crank backup

Operates on A.C. mains as well as by hand cranking. A centrifugal clutch (governor) holds the output voltage constant in hand mode, and the generator handle is collapsible.

05 TECHNICAL SPECIFICATIONS

Drive	**Mains motor (230 V, 50 Hz A.C.) plus collapsible hand-crank backup**
Body	**Seasoned wooden cabinet with sunmica finish, detachable lid, handle, lock**
Scale Length	**100 mm (approx.)**
Accuracy	**As per IS: 2992-1987**
High Voltage Test	**Min. 2 kV or (rated voltage + 1) kV A.C. (r.m.s.), 1 min**
Insulation Resistance Test	**Better than 50 MΩ at 500 V D.C.**
Guard Terminal & Infinity Adjuster	**Guard ring and guard terminal eliminate surface-leakage errors; externally accessible infinity adjuster provided**
Dimensions (incl. top cover, approx.)	**14 × 10 × 10 inch**
Weight (incl. top cover)	**12 kg (approx.)**
Response Time	**3 seconds**

Ranges are listed on the following pages. Specifications are approximate where marked.

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06 STANDARD RANGES BY RATED VOLTAGE

RATED VOLTAGE	INSULATION RESISTANCE RANGE (MΩ)
500 V	0–1,000
1,000 V	0–2,000
2,500 V	0–5,000 · 0–10,000 · 0–20,000
5,000 V	0–5,000 · 0–10,000 · 0–20,000 · 50–50,000 · 100–100,000
10,000 V	0–10,000 · 0–20,000 · 50–50,000 · 100–100,000

Ranges shown are the standard builds. Any other range can be manufactured against specific requirements.

07 COMBINING VOLTAGES AND RANGES

Dual voltage & dual insulation range

Two rated voltages, each with its own range. The selected range must lie within the rated resistance listed for that voltage. Not available for 10,000 V.

Single voltage, dual insulation range

One rated voltage with two insulation ranges within the listed rated resistance for that voltage. Not available for 10,000 V.

Dual voltage, single insulation range

Two rated voltages sharing one insulation range within the listed rated resistance. Not available for 10,000 V.

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08 MULTIRANGE SERIES: CIE/777 HM/MR

SERIES	RATED VOLTAGE	INNER SCALE (MΩ)	OUTER SCALE (MΩ)
MR-A	5,000 V	0–5,000	50–100,000
	2,500 V	0–2,500	25–50,000
	1,000 V	0–1,000	10–20,000
MR-B	5,000 V	0–2,000	10–10,000
	2,500 V	0–1,000	5–5,000
	1,000 V	0–500	2–2,000
MR-C	5,000 V	100–100,000	Single scale
	2,500 V	50–50,000	Single scale
MR-D	5,000 V	0–5,000	50–100,000
	2,500 V	0–2,500	25–50,000
MR-E	5,000 V	50–50,000	Single scale
	2,500 V	25–25,000	Single scale
	1,000 V	10–10,000	Single scale

Multirange instruments carry a range selector switch, so one instrument covers every voltage and range listed in its series. Any other configuration can be built to order.

09 SUPPLIED AND OPTIONAL

Supplied as standard

- ✓ Mains cord, 2 m
- ✓ 1 pair of H.V. testing leads

Optional, at extra cost

- Rexine carrying case
- 15-minute timer facility
- Testing leads, 1 m or 2 m
- Additional / special-grade H.V. testing leads, 2 m

10 HOW TO ORDER

To order, quote the **model**, the **rated voltage** and the **insulation range**, for example **CIE/777 HM, 500 V / 0–1,000 MΩ**. Any range not listed can be manufactured against specific requirements.

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